2220 - Entrepreneurial Finance and Venture Capital

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Lecture #9

Where we are

- Evaluating investment opportunities
  - Overall framework: People, Opportunity, Context
  - Financial implications of business models: 3-stage model
  - Valuation: VC method vs standard DCF
  - Experimentation, real options and multistage finance: decision trees
- **Assessing financing alternatives**
  - Deal structure and terms
  - Angels, accelerators, venture capital
- Realizing returns
  - Selling the venture vs IPO vs staying private

Financing decisions

- Modigliani-Miller: irrelevant, just different ways of slicing the pie
  - Assumptions: no taxes, no transaction costs, symmetric information
- Corporate finance: focus on trade-off between
  - Interest tax shields
  - Costs of financial distress
- Entrepreneurial finance: asymmetric information plays the key role
  - Adverse selection
  - Moral hazard

Asymmetric information

- Before deal, insiders (founders and existing investors) know more than outsiders (new investors)
  - How skilled and committed are the founders?
  - How reliable is the product? How well will it scale?

  $\rightarrow$ Adverse selection: insiders can exploit information advantage when negotiating deal

- After deal, investors cannot observe everything founders do
  - How hard are they working?
  - What risks are they taking?

  $\rightarrow$ Moral hazard: founders do not always act in best interest of investors

Adverse selection example

- Suppose there are two types of startups, Good and Bad
  - Both need to raise 5M of upfront investment from VCs
  - After investment, Good startups are worth 10M and Bad startups are worth 2M
- Founders know if their startup is Good or Bad, but VCs don't
  - VCs do know from experience that 10% of startups are Good
- Bad startup founders have incentive to hide their quality: if they convince VC to invest they will get a share of the 2M
- Will the VC invest?
  - Expected NPV $= 10 \times 10\% + 2 \times 90\% - 5 = -2.2$
  - No investment, even though Good startups are NPV>0 projects
  - Note adverse selection $\neq$ uncertainty: founders know quality of their startups

Inc.

The Magazine for Growing Companies

HOW TO BUILD A COMPANY TO LAST 100 YEARS

PAGE 32

THE ART OF GIVING (AND TAKING) EQUITY

PAGE 120

THE NEXT STEVE JOBS

8 WOMEN WHO COULD OWN THE FUTURE

PAGE 47

Elizabeth Holmes has taken Theranos to a $10 billion valuation in 12 years.

+

DO YOU *REALLY*

Moral hazard example

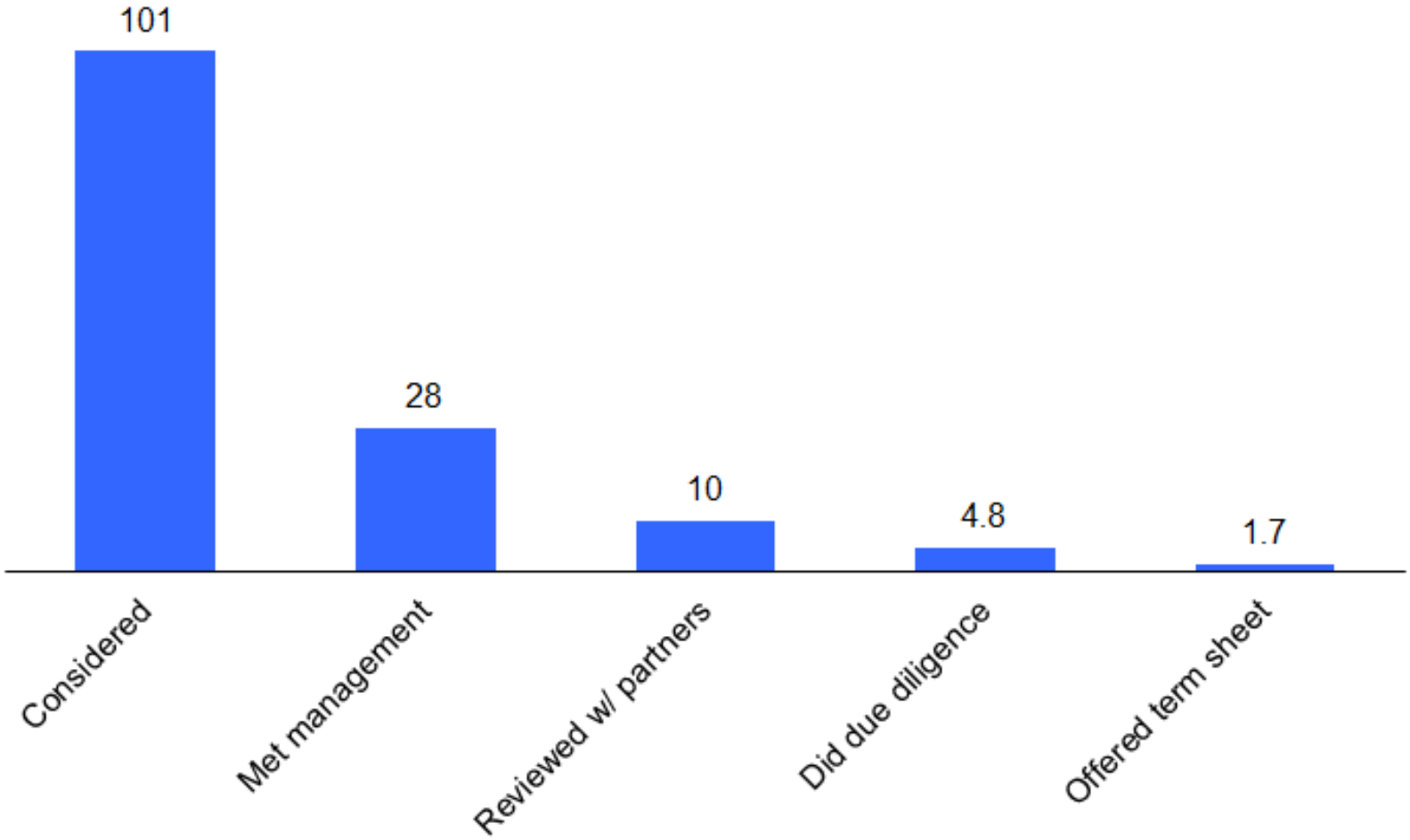
- Suppose a startup needs to raise 5M of upfront investment from a VC
- Both founders and VC know that
  - If founders work hard the expected value of the startup will be 8M post-money
  - If founders don't work hard, the expected value of the startup will be 3M post-money
- However, VC cannot observe if founders are working hard
- Suppose VC offers to buy 100% of the equity and pay founders a fixed salary. Will founders work hard?
  - No matter how high the salary, VC receives 100% of the value created by hard work, while founders put in 100% of the effort
  - VC and founder interests not aligned
- Same problem exists with VC value-added and effort

FYRE

How VCs address adverse selection and moral hazard

1. Screening
   - Distinguish Good startups from Bad ones
2. Monitoring and control
   - Obtain information on ongoing performance and activities
   - Ensure control over key decisions where interests might be misaligned
3. Incentives
   - Align interests between founders and VCs

Screening: potential investments per closed deal



Source: Gompers et al. 2016. How Do Venture Capitalists Make Decisions? National Bureau of Economic Research Working Paper

Due diligence

- Founder references
- Business: market size, value proposition, business model
- Technical: technology, intellectual property
- Legal/accounting: contracts, financial statements
- Average deal takes 83 days to close, 118 hours of due diligence, 10 reference calls (Gompers et al., 2016)

Staging

Staging plays multiple roles in entrepreneurial finance:

1. Maximizes value of abandonment options
2. Enables investors to screen
   - Good startups will prefer to defer raising funds until investors learn about their quality, at which point they will get better terms
   - Bad startups have the opposite incentive
3. Acts as incentive/control mechanism
   - Keeps founders on "tight leash": meet milestones or investors can pull plug

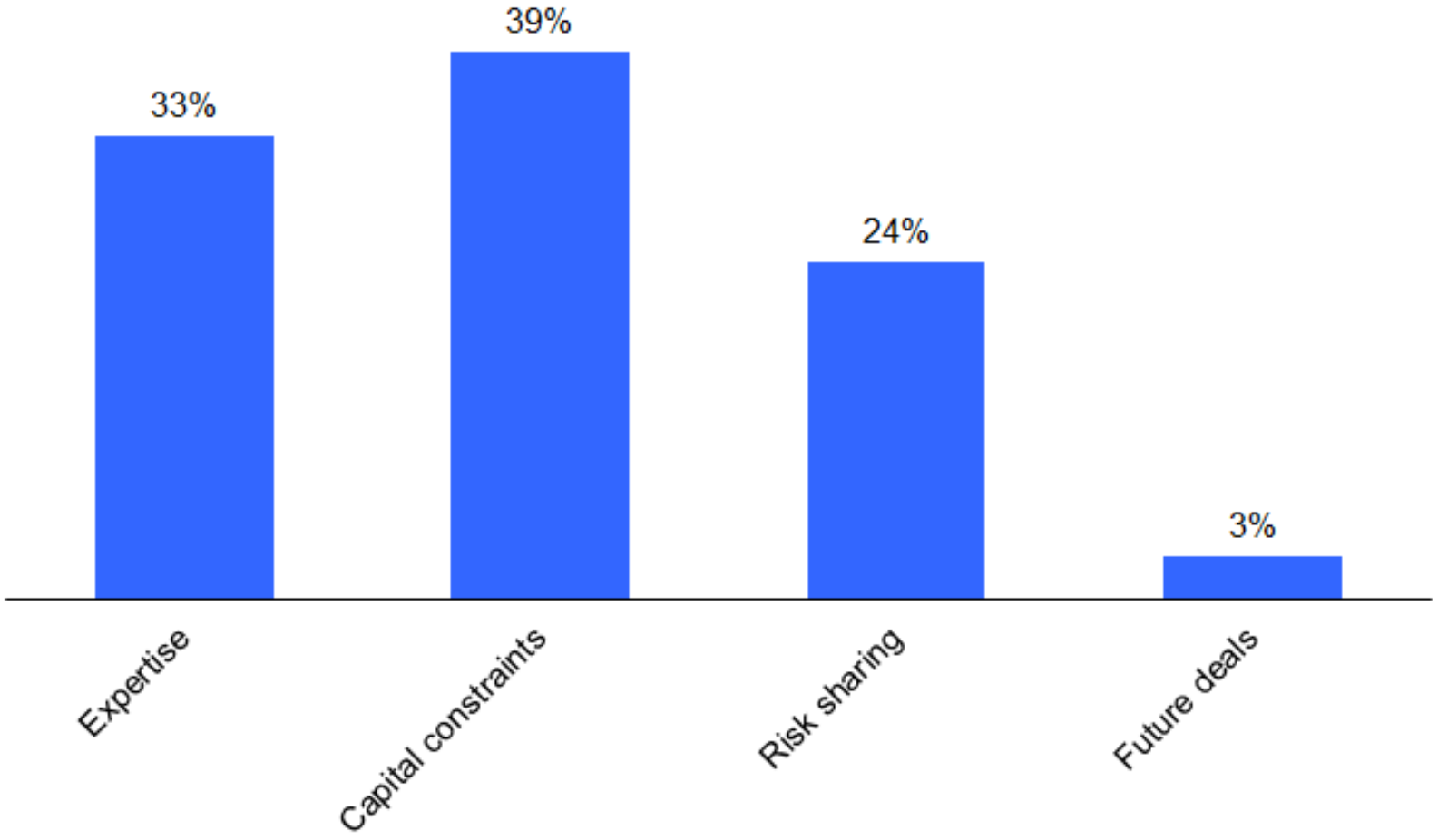
Contingent control rights

- Control rights are often contingent on performance
  - If firm performs well, founder retains/increases control
  - If firm performs poorly, control shifts to VC, who can replace management
- This enables investors to screen
  - Good startups are more likely to accept, since they expect to perform well
- Also strengthens founder incentives to perform
- How is this implemented in practice? Examples:
  - Voting control shifts to VC if EBIT$<$ target
  - VC gets board majority if startup fails to recruit CEO within 9 months
  - Founders have veto over sale if revenues and profits $\geq$ 80% of target

Syndication

- VCs typically invest alongside other VCs ($\approx$ 65% of deals). Why?
- They do so for several reasons
  - To get a second opinion and improve screening, especially in early stage
  - To provide complementary expertise after deal is signed
  - To overcome capital constraints
  - To diversify risk
  - To gain access to future deals

Most important factor for syndication



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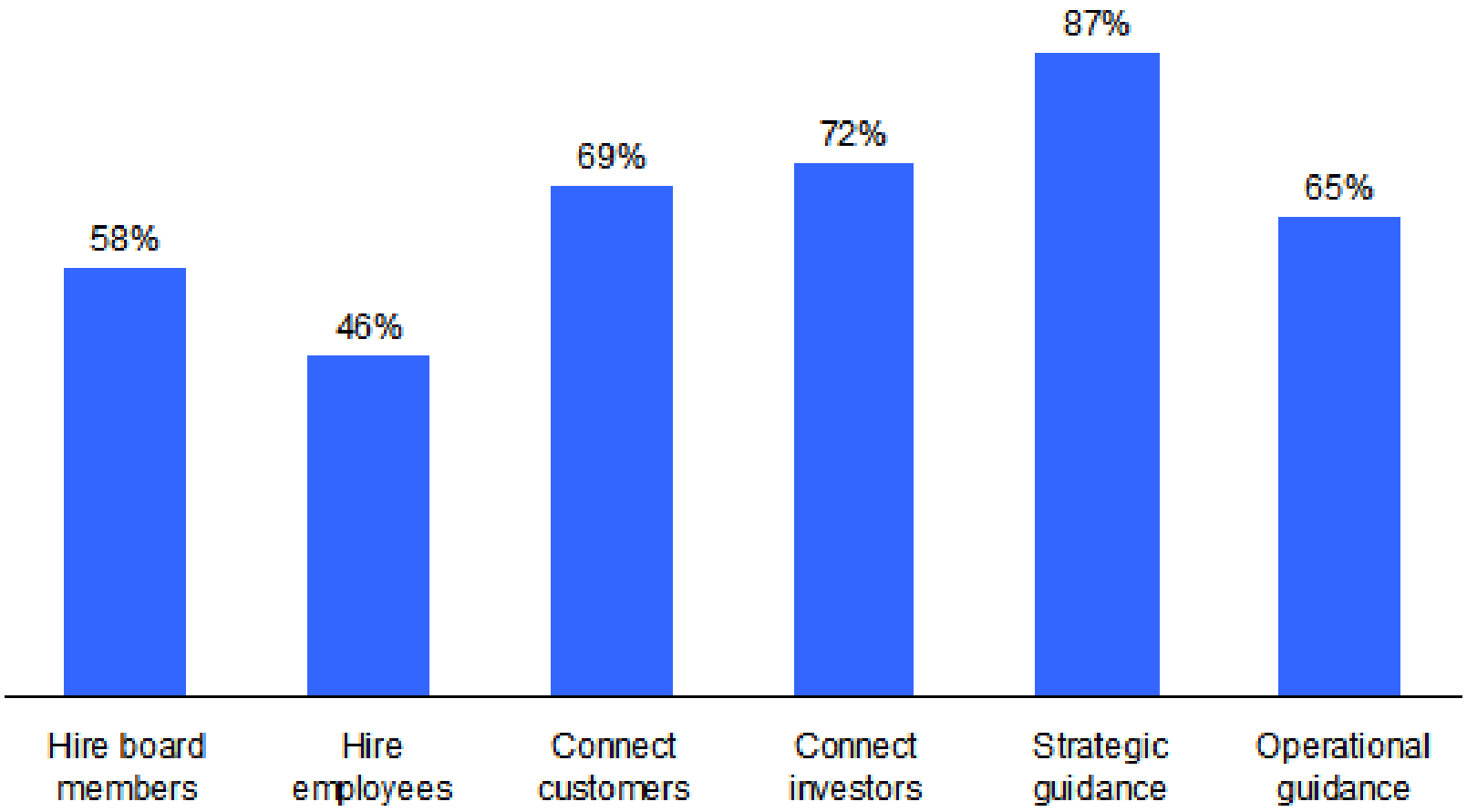
Monitoring & control

- Frequent informal interactions
- Board representation
- Voting rights: key decisions may be subject to shareholder vote
  - Major acquisitions
  - Asset sales
  - Subsequent financings
  - Election of directors
- Class voting rights: specific VC approval may be required for
  - Sale, merger or liquidation
  - Changes in corporate charters
  - Major acquisitions or budget changes
  - Major management changes (e.g. CEO)

VC activities in portfolio companies



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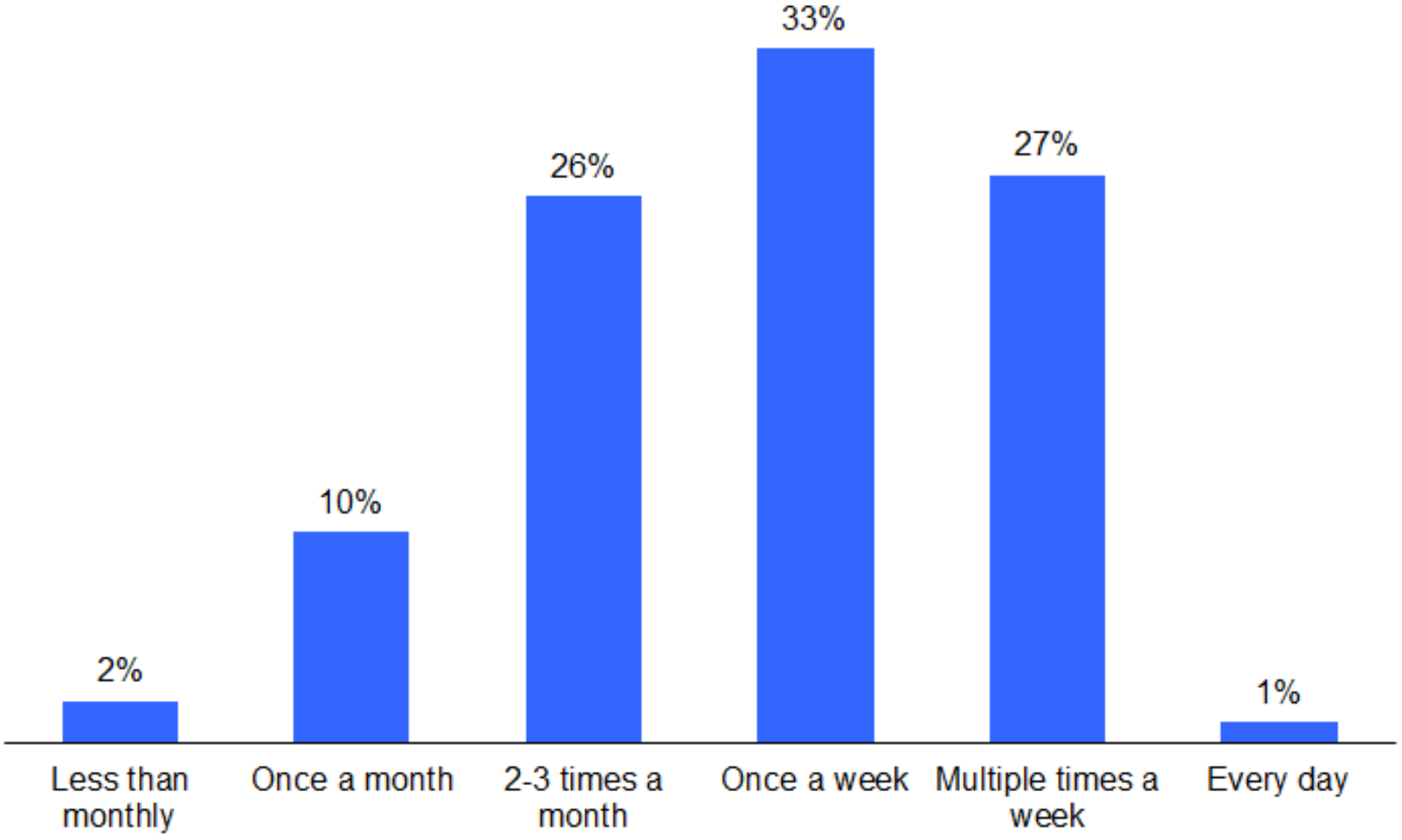
Frequency of interaction



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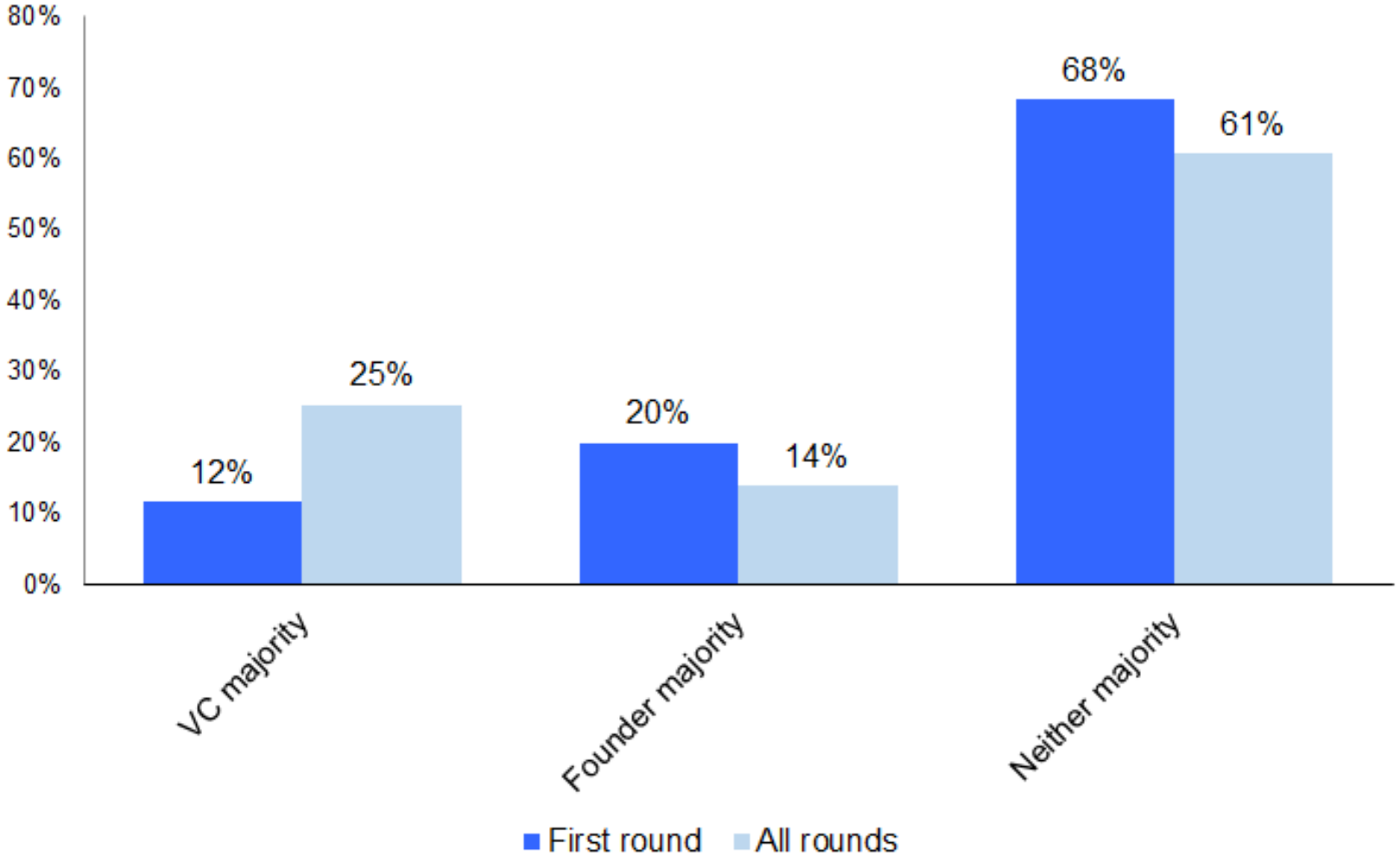
Board of directors

- Formal board typically created once outside investors come in
- Size: 3-6 in early stages, 5-8 in later stages
- Meeting frequency: from quarterly to monthly, more often in early stages
- Board members: founders, investors, independents
  - Lead investor for each round usually demands one board seat, at least
  - Founders often lose majority with series A round
  - Common for independent directors to have deciding vote when founders and investors disagree

The dual role of the board

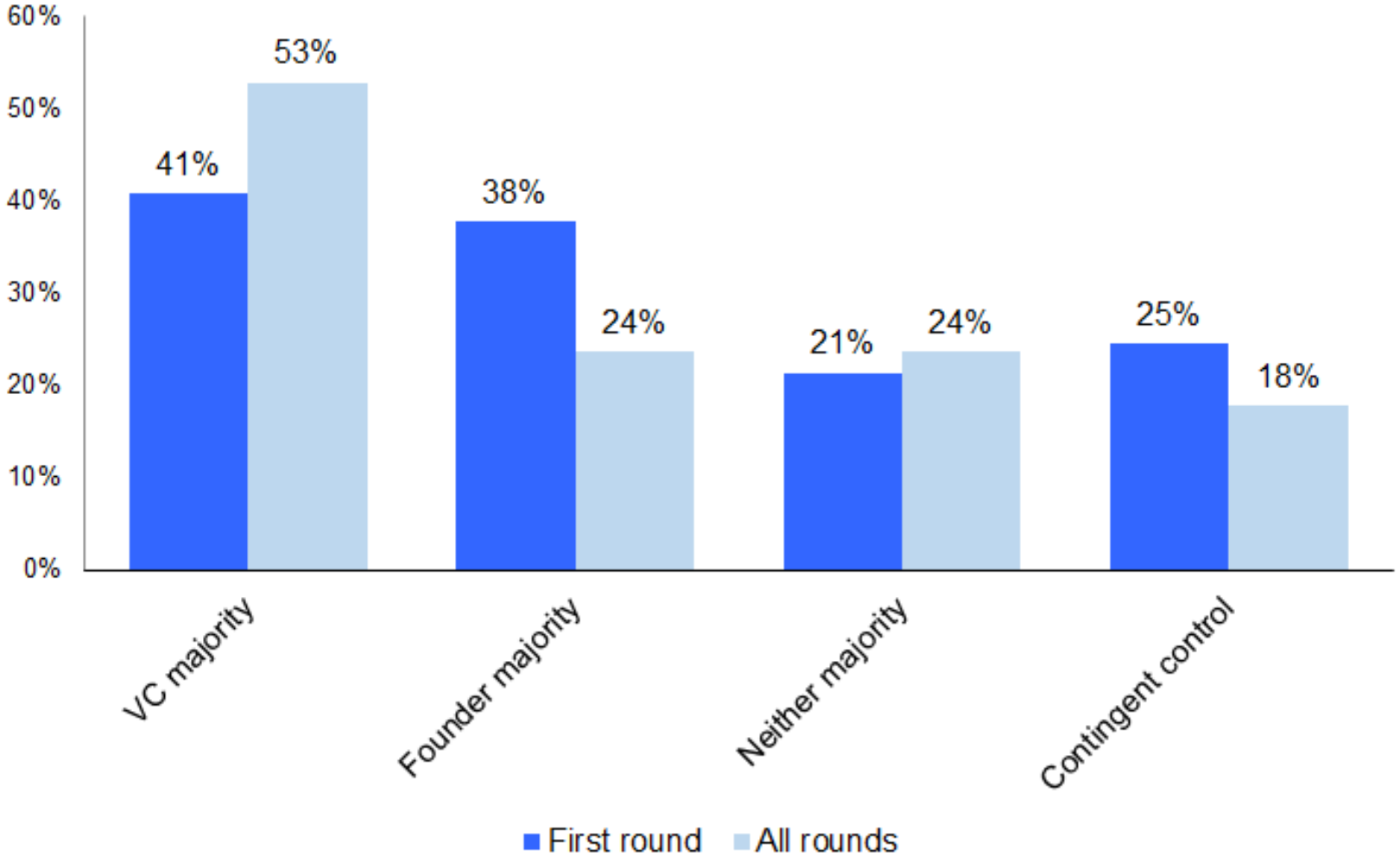
- Advising the CEO/founding team
  - Ratify general corporate strategies and decisions
  - Provide feedback and advice
  - Complement the team's skills
- Evaluating the CEO/founding team
  - Power to replace CEO
  - Set compensation
  - Audit financials
  - Approve major decisions, e.g. budget (depending on deal terms)
- Ultimately, board represents interests of shareholders, not founders

Board control



Source: Kaplan and Stromberg 2002. Financial contracting theory meets the real world: An empirical analysis of venture capital contracts

Voting control



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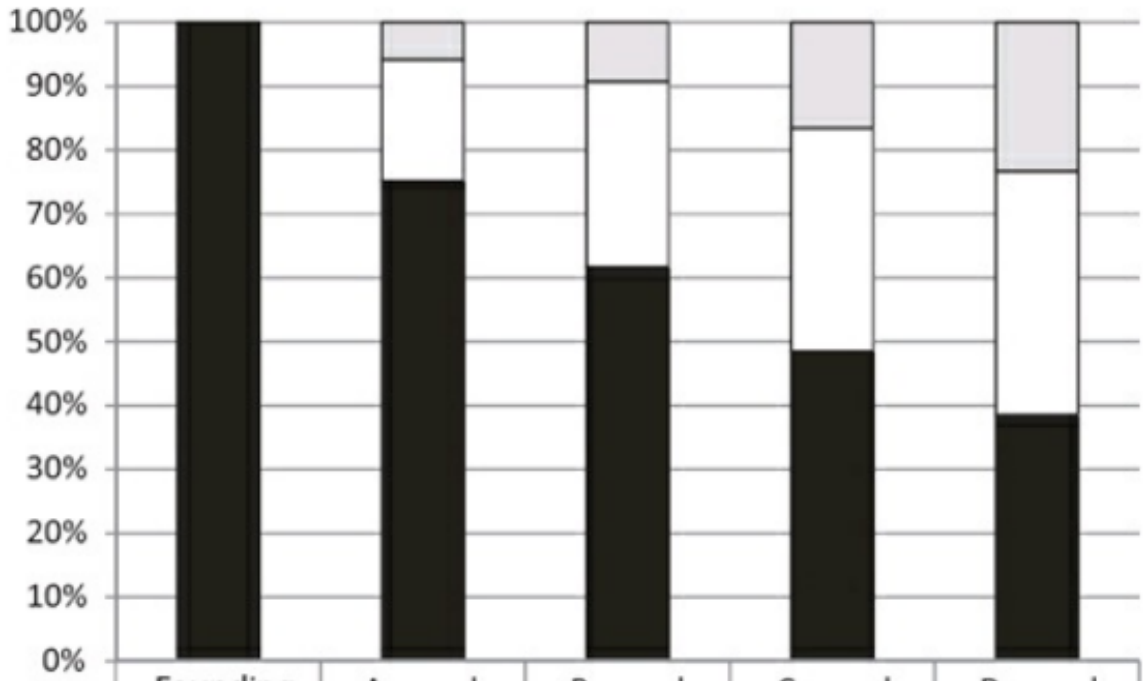
Startups led by founder-CEO at each round



	Founding	A-round	B-round	C-round	D-round
On 3rd (or more) CEO	0%	6%	9%	17%	23%
On 2nd CEO	0%	19%	29%	35%	38%
Founder still CEO	100%	75%	62%	48%	39%

Figure 10.1. Percentage of Startups Still Being Led by the Founder-CEO

Source: Wasserman, Noam. The founder's dilemmas: Anticipating and avoiding the pitfalls that can sink a startup. Princeton University Press, 2012

Incentives

- Founder equity split
- Vesting
- Choice of security
- Pro-rata rights
- Anti-dilution provisions
- Exit incentives
- Non-compete agreements: founders cannot join competitor for, e.g., 2 years
- Stock options: key hires are awarded options to buy stock in company

Founder equity split

- Why do VCs care?
  - Ensure key team members are invested in success of venture

- Two main views

1. Split according to expected future contributions
   - Maximize incentives for most important roles, e.g. CEO

2. Split equally
   - Avoid tensions, preserve fairness
   - Forces hard thinking about who should be a founder
   - *"If you aren't willing to give your partner an equal share, then perhaps you are choosing the wrong partner"* Michael Seibel, YC

Difference between largest and smallest equity stake

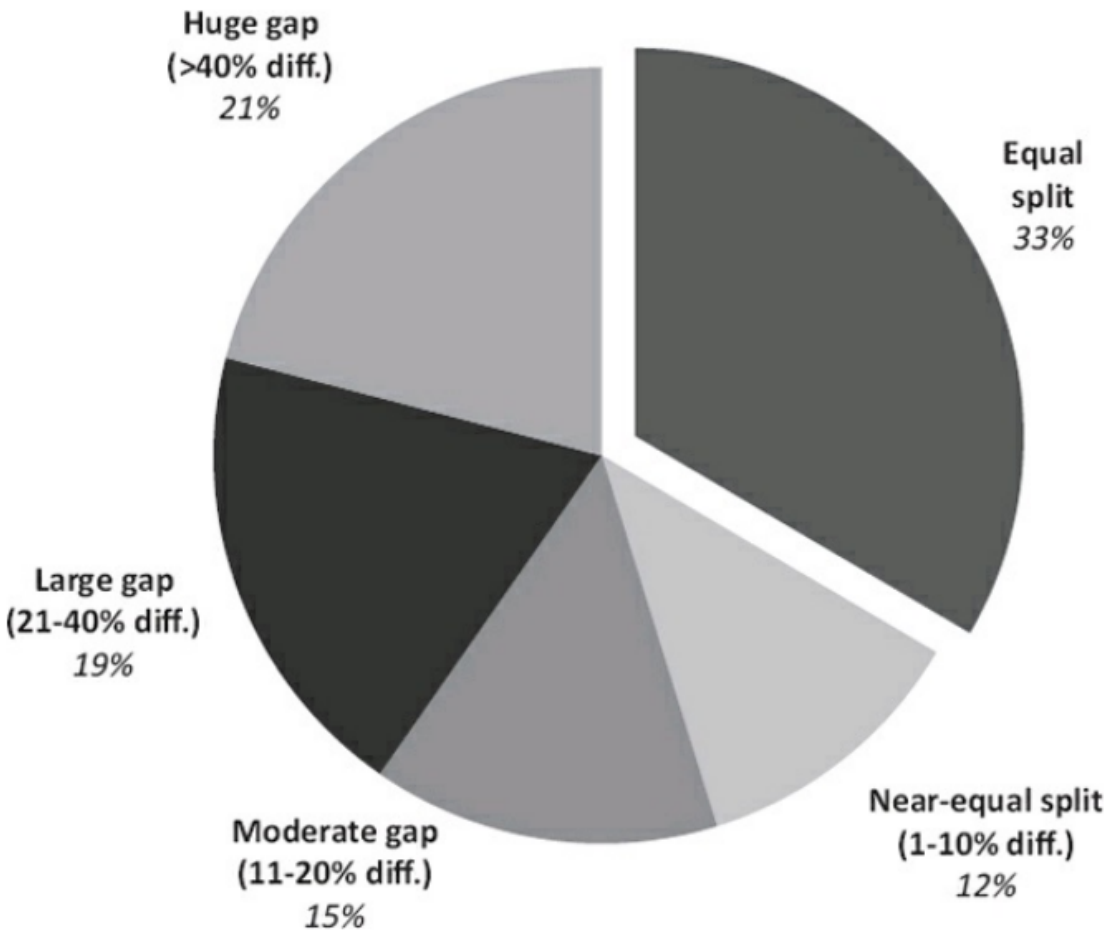


Figure 6.4. Difference in Equity Stakes between Founder with Largest Stake and Founder with Smallest Stake

Source: Wasserman, Noam. The founder's dilemmas: Anticipating and avoiding the pitfalls that can sink a startup. Princeton University Press, 2012

Vesting

- Founders only earn their shares if they stay with company
  - e.g. 25% after first year, then monthly increments over following 3 years

- Zipcar is a car-sharing company founded by Robin Chase and Antje Danielson in 2000
- The founders shook hands on a 50/50 equity split early on, without vesting
- But Danielson never quit her job, and Chase built the company much on her own
- In 2013, Zipcar was sold to Avis for $491 million
- Chase later reflected "That was a really stupid handshake..."

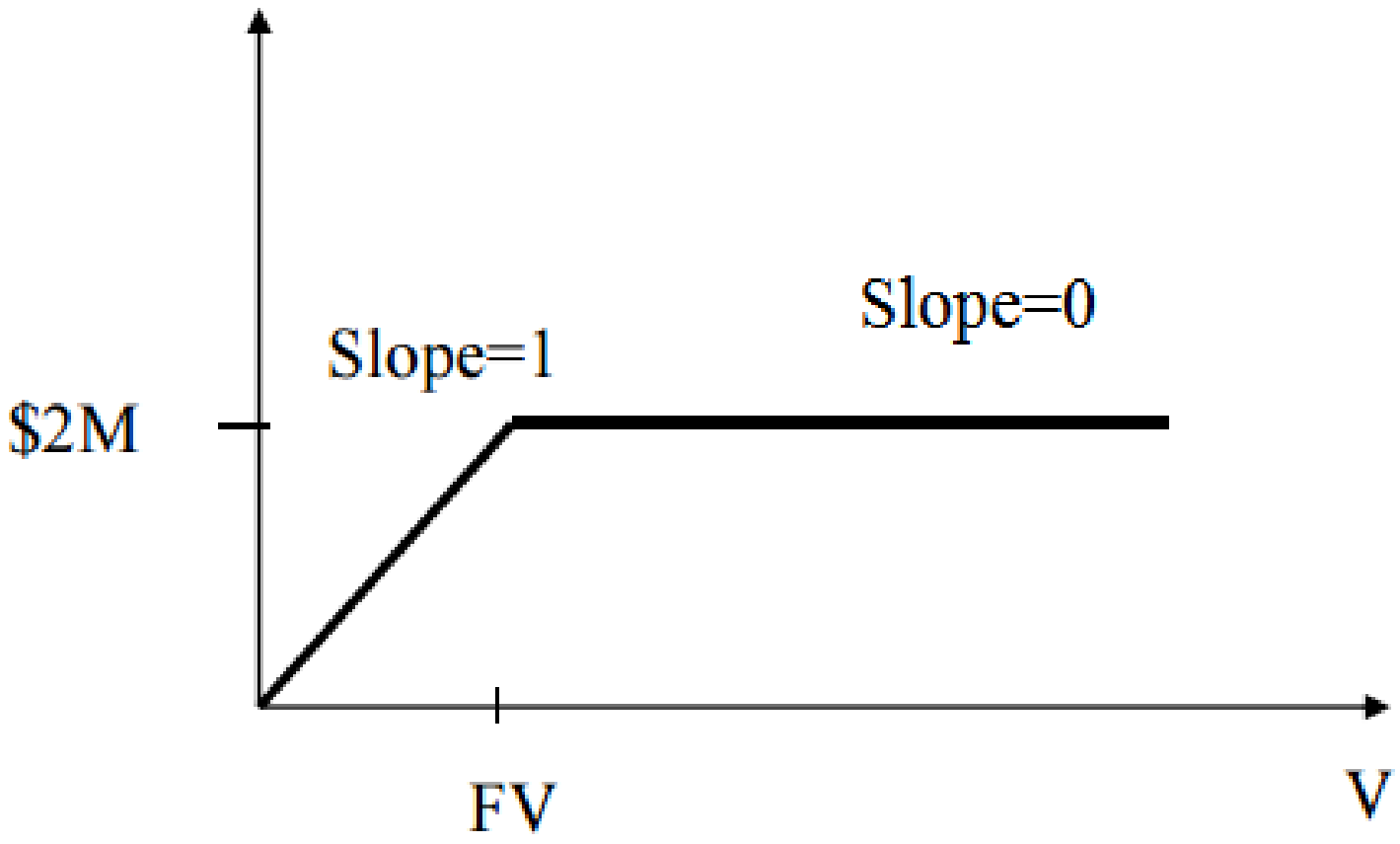
Choice of security

- VCs typically do not get common stock
- Deals tend to take one of two forms
  1. Convertible preferred stock (CP)
  2. Participating convertible preferred stock (PCP)
- Usage is split about 50/50 (Gompers et al., 2016)

Preferred stock

- Preferred stock behaves much like debt
- It is a fixed claim: face value plus possibly a pre-specified dividend
- It enjoys a **liquidation preference** over common stock: face value and accrued dividends get paid first in event of sale or liquidation
  - Face value can be equal to investment amount (1X liquidation preference)
  - It can also be a multiple of that amount (2X, 3X, etc)

Investor payoff diagram: preferred stock



- Example: preferred stock with face value (FV) of 2M
- VC enjoys downside protection through liquidation preference, but
  - No upside
  - No incentives to add value

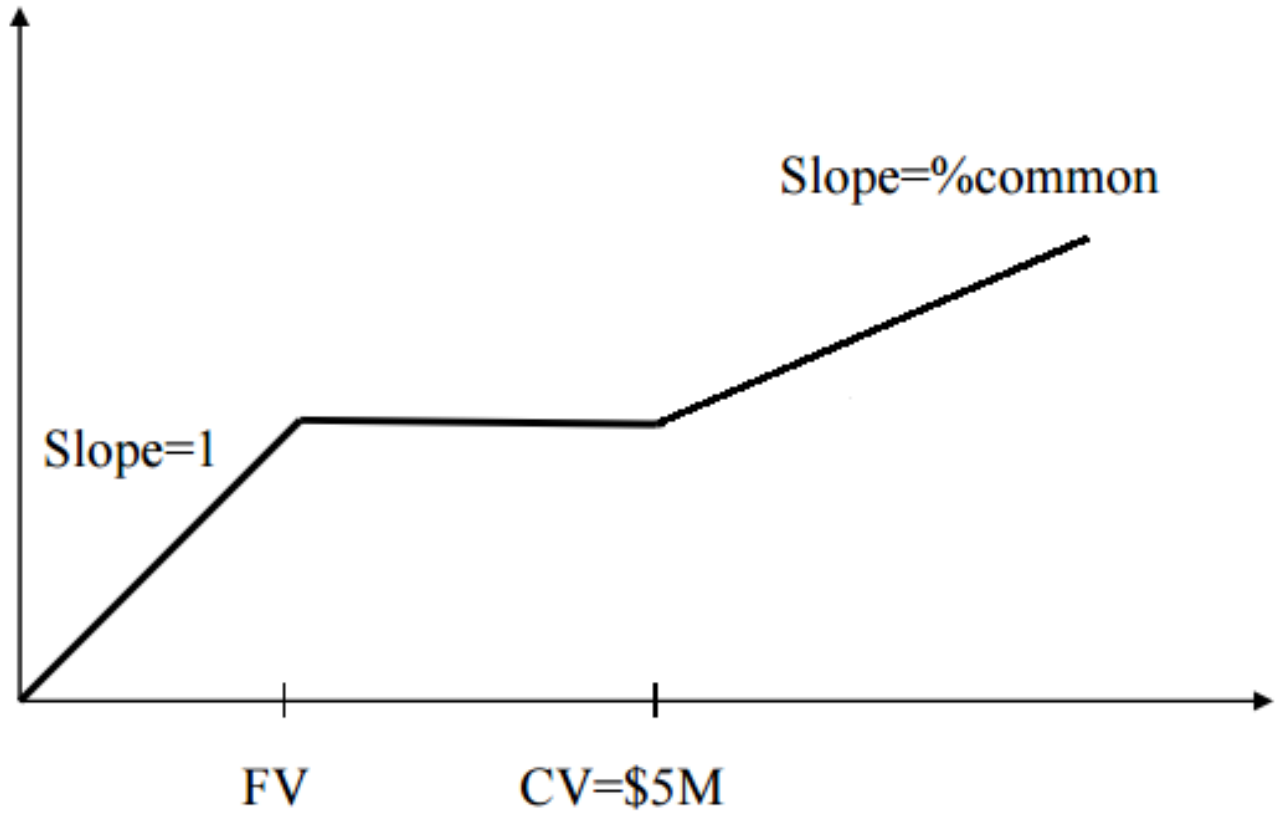
Convertible preferred stock (CP)

- Depending on exit value, investors can choose to either
  - Keep preferred stock and receive face value
  - Convert to common stock at a pre-specified price
- Optimal to convert if value of converted common stock exceeds face value of preferred
- Note: most contracts include automatic conversion at IPO provided IPO price and proceeds are high enough
  - Underwriters usually demand all preferred stock be converted upon IPO
  - Automatic conversion avoids negotiation with preferred investors

Convertible preferred example

- If initial investment is 2M and conversion price is 5/share, VC can convert into 400,000 shares
- If there are 600,000 common shares outstanding before conversion, then own 40% upon conversion
- Ignoring dividends, investor will convert if 40%*V>2M or V>5M

Convertible preferred payoff



FV: Face value of preferred stock
CV: Min. enterprise value at conversion

- VC and founders share upside
  - VC is compensated for capital invested and risk taken
  - Incentives for both founders and VC to create value
- VC enjoys downside protection through liquidation preference

Why not just use common stock?

- Suppose a VC invests 2M under 5M post-money valuation
- The day after, the founder gets an offer to sell the firm for 2.5M
- With common stock
  - Founder gets $3/5 \times 2.5 = 1.5$M without putting in *any* funding or work
  - VC gets $2/5 \times 2.5 - 2 = -1$M
- With preferred stock
  - Liquidation preference ensures VC gets the 2M back before founder gets anything
- Preferred stock eliminates incentive to "take the money and run"
- Other ways to mitigate problem: vesting, non-competes, class voting

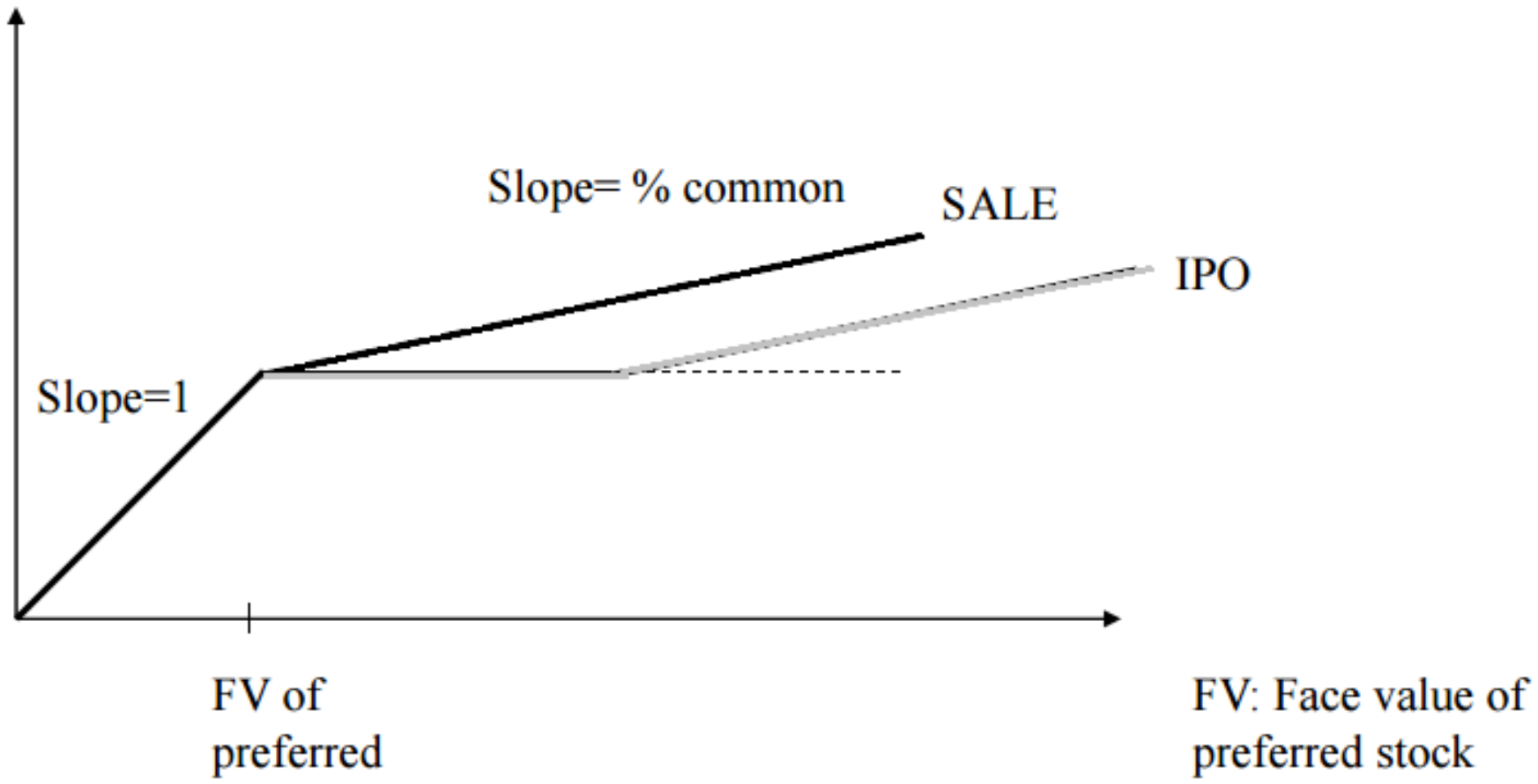
Participating convertible preferred stock (PCP)

- CP with extra feature: "in the event of liquidation or sale" investor gets face value *and* equity as if stock were converted

- Equal to holding simple preferred + common if firm is sold or liquidated
  - In previous example, investor would get 2M *and* 40% of common stock
  - Participation could be capped: e.g. investor gets up to 4X original investment amount

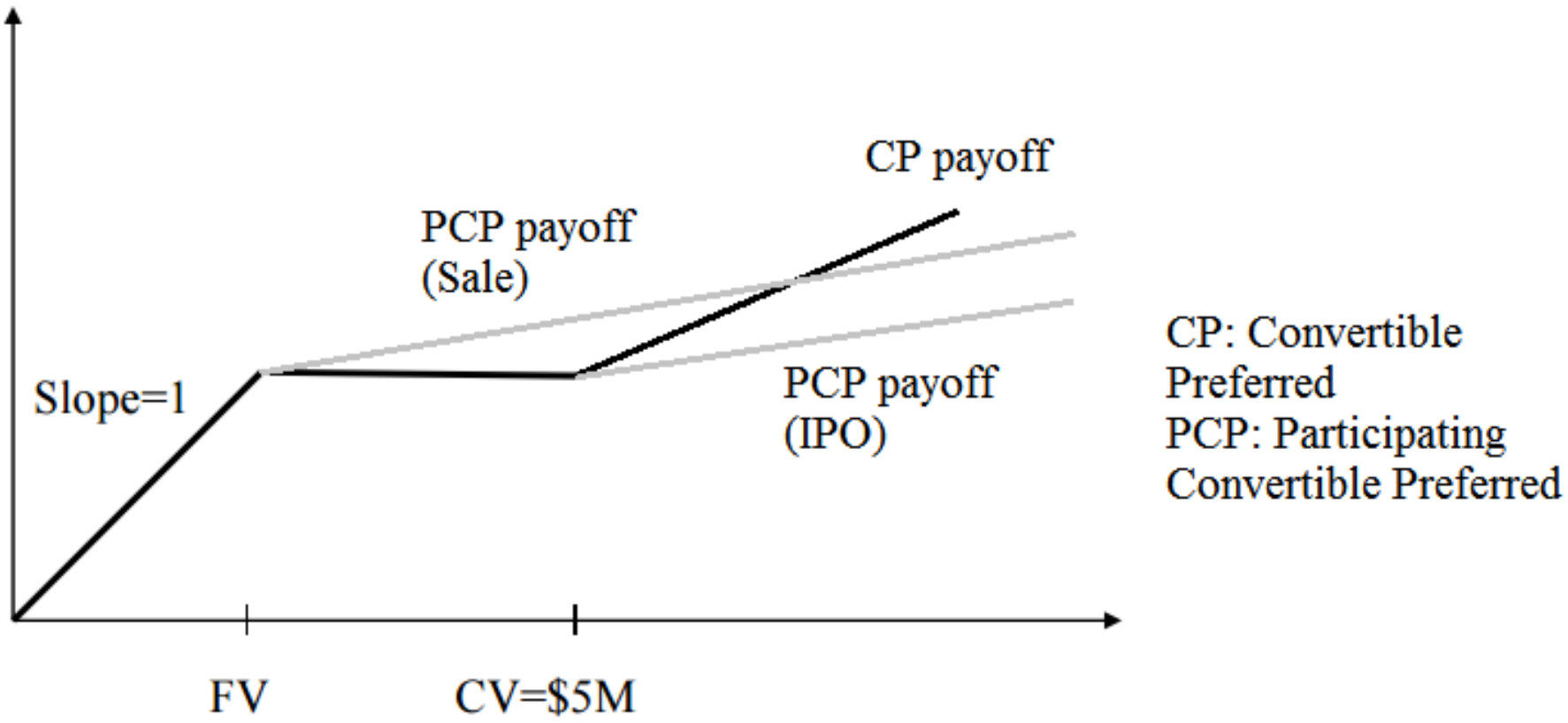
- Automatic conversion to common at IPO still applies
  - Payoff equal to convertible preferred if firm exits through IPO

Participating convertible preferred payoff



- PCP strictly better than CP for VC holding stake constant
- Founders might push for lower VC stake/higher valuation in return for PCP

CP vs PCP payoff if participation is offset by lower stake



- In this case, PCP offers
  - Stronger downside protection, smaller upside for VC
  - Stronger incentives for founders, weaker incentives for VC

Pro-rata rights

- Right to participate in future rounds and maintain ownership %
- Ensures early stage investors can keep investing in their most successful deals
- Prevents founders and later stage investors from taking advantage of early stage investors
  - Suppose a startup is raising a series A round
  - Founders and series A investor agree on low valuation in order to dilute seed investors
  - Founders get compensated for also being diluted, e.g. through stock options
  - Pro-rata rights enable seed investors to invest under same terms and avoid this

Anti-dilution protection

- When firm raises funding at a price below a prior round's price (down-round), prior round's conversion price is lowered
- This protects prior round's investors from dilution
  - Specified in nearly all deals
  - Founders and unprotected investors bear the cost of protection
- Strengthens incentives for founders to work hard and meet milestones

Types of anti-dilution protection

- Full ratchet anti-dilution: conversion price is lowered to price of new round, regardless of the new round's size

- Weighted average anti-dilution: new conversion price takes into account the size of the new round
  - New conversion price = [(A+C)/(A+D)]*old conversion price
    - A: # of shares outstanding before new round
    - C: # of shares issued in new round if it had been raised at the old price
    - D: # of shares actually issued in new round
  - The larger the new round, the stronger the protection

Anti-dilution example

- Suppose firm has 2M shares outstanding
  - 1M shares of founder common stock
  - €1M in convertible preferred for a VC, with conversion price of €1
  - Upon conversion, VC would get $1/2 = 50\%$ of equity
- Firm issues 50,000 shares at €0.50 in new round, raising €25,000
- Full-ratchet
  - New conversion price is €0.50
  - VC can now convert to 1M/0.5=2M shares or 2/3.05=65.57% of equity
- Weighted average
  - New conversion price is [(2+0.025)/(2+0.05)]*1=€0.988
  - VC can now convert to 1M/0.988=1.012M shares or 1.012/2.062=49.08% of equity

Exit incentives

- Redemption rights: preferred stock is often redeemable
- Redeemable preferred allows investor to recover face value and dividends after a given term
  - Meant to force exit and prevent "life-style company"
  - Term is usually 5 to 8 years
  - If firm cannot redeem, penalties kick in, e.g. increased board seats for VC
- Other tools to align exit incentives
  - Tag along: if offer is made for one shareholder's shares, offer must be extended to other shareholders
  - Drag along: if majority of board/shareholders approve offer, all shareholders forced to sell

Summing up

- The key factor shaping VC deals is asymmetric information, namely adverse selection and moral hazard
- VCs use a variety of tools to
  - Screen high potential startups
  - Monitor and control performance
  - Create incentives for founders and VCs to maximize value
- Deal terms should be understood/evaluated in light of these objectives